

Testing REST API Using Postman

*A practical guide to GET, POST, PATCH, and DELETE operations
using Postman and Supabase*

Version 1.0

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May 03, 2026

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Introduction

This document provides an overview of the REST API endpoints built on the Students database, created in Supabase and tested using Postman.

The API allows you to interact with the students database through standard HTTP methods — GET, POST, PATCH, and DELETE.

Use this document to learn how to use the API methods using Postman.

Scope

The Student Database used in this document was created in Supabase for testing and demonstration purposes only. The data present in the database does not represent any real student information. Postman tests and demonstrates the API requests documented in this guide.

Endpoint reference

Base URL: <https://etyucsgtbtmxmlimthorwr.supabase.co/rest/v1/>

Endpoint: students

Full endpoint URL:

<https://etyucsgtbtmxmlimthorwr.supabase.co/rest/v1/students>

Authentication

Authenticate your calls to the Student Database. Include these headers in every Postman request.

Table 1. Headers required on every Supabase API request from Postman

Key	Value	Description
apikey	<your-anon-key>	Your Supabase anon public key
Authorization	Bearer <your-anon-key>	Bearer token for authentication
Content-Type	application/json	Request format

Note:

To find your anon key, go to your **Supabase Dashboard > Project Settings > API > Project API Keys** and copy the **anon / public** key. Use the API key only through the **Headers** tab in Postman. Do not use it as a query parameter in the URL.

HTTP methods overview

Table 2. Supported HTTP methods and CRUD actions

Method	Action
GET	Retrieve student records
POST	Create a new student record
PATCH	Update an existing student record
DELETE	Delete a student record

Note:

The API also supports the **PUT** method. **PUT** replaces the entire record. Use **PATCH** when you want to update specific fields only.

Setting up Postman

1. Create a new request

Select the + icon on top and select HTTP.

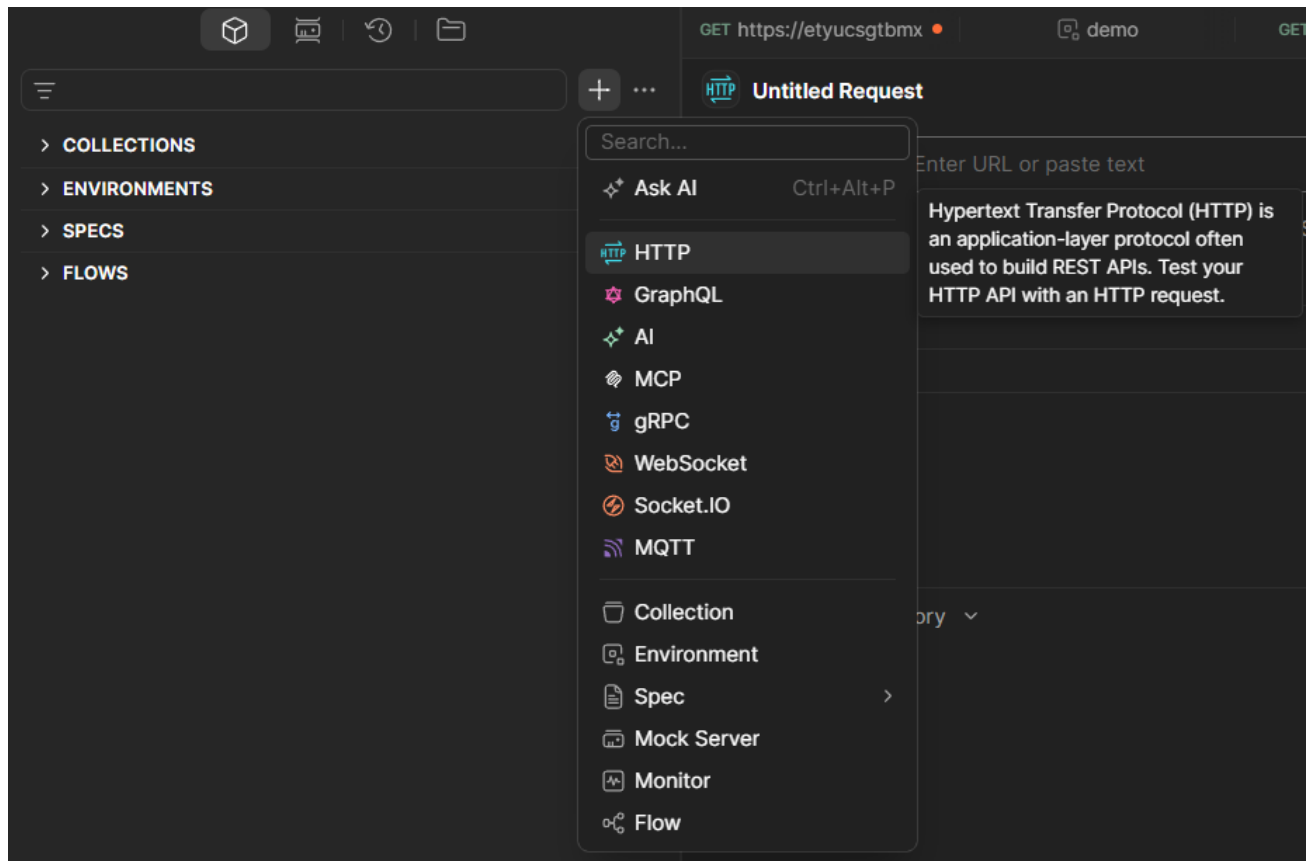


Figure 1. New request type selection menu in Postman

2. Select the HTTP method

At the top-left of the request panel there is a dropdown menu, it is usually set to **GET** by default. If you need to use **POST**, **PATCH**, or **DELETE**, select the dropdown and select the required method.

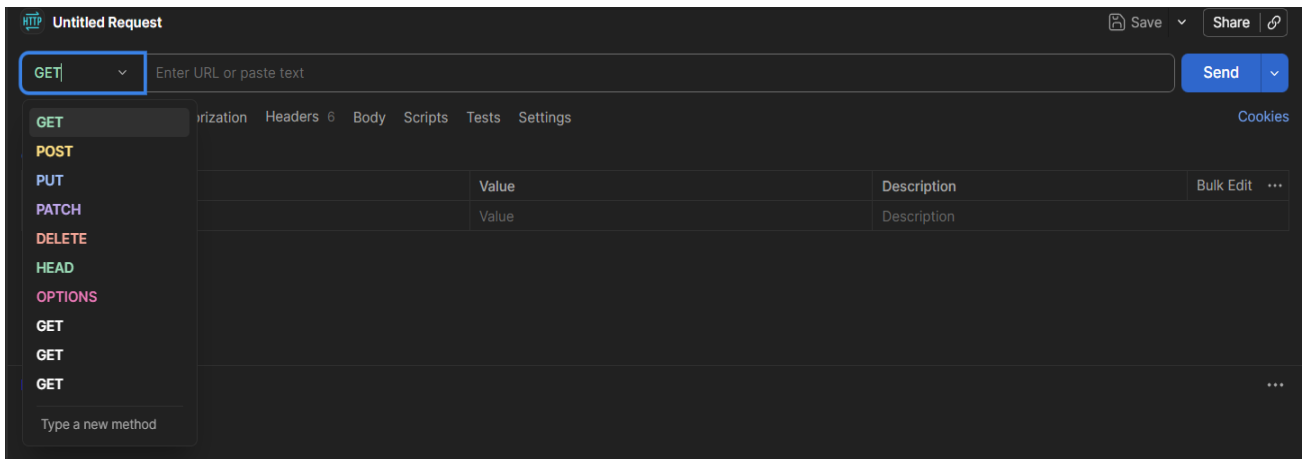


Figure 2. HTTP method dropdown showing available methods

3. Enter the endpoint URL

In the **URL text box**, enter the full endpoint URL.

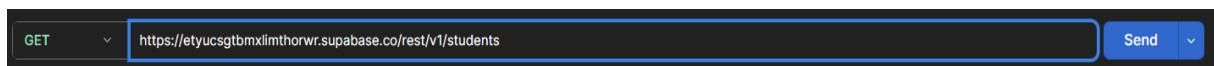


Figure 3. Endpoint URL entered in the URL text box

4. Add authentication headers

Below the **URL text box**, select the **Headers** tab and enter the authentication headers as specified in the [Authentication](#) section.

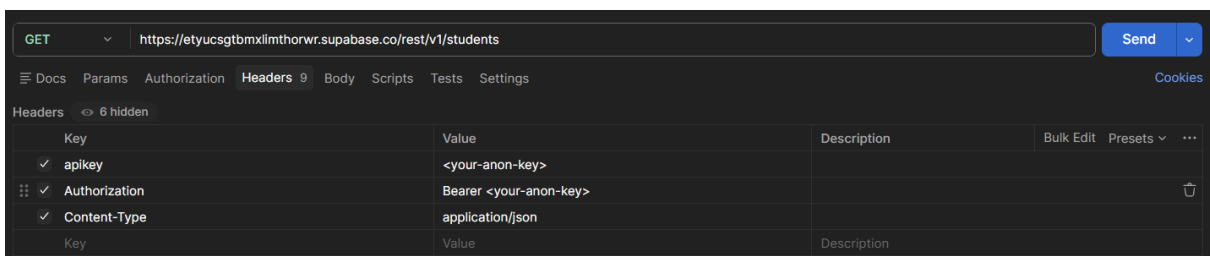


Figure 4. Authentication headers added in the Headers tab

5. Select **Send**.

GET method

The GET method only reads data and does not modify any records. Use this method when you want to fetch student details from the database.

Endpoint: **https://etyucsgtbtmxmlimthorwr.supabase.co/rest/v1/students**

1. Enter the endpoint URL in the URL **text box**.

2. Select **GET** from the HTTP method dropdown.

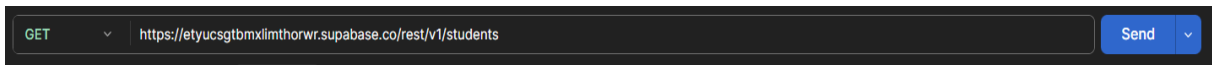


Figure 5. GET request URL in the URL text box

3. Select the **Headers** tab and add the required authentication headers as specified in the [Authentication](#) section.
4. Select **Send**.
5. Verify that the response shows a **200 OK** status and student records in the **Body** tab.

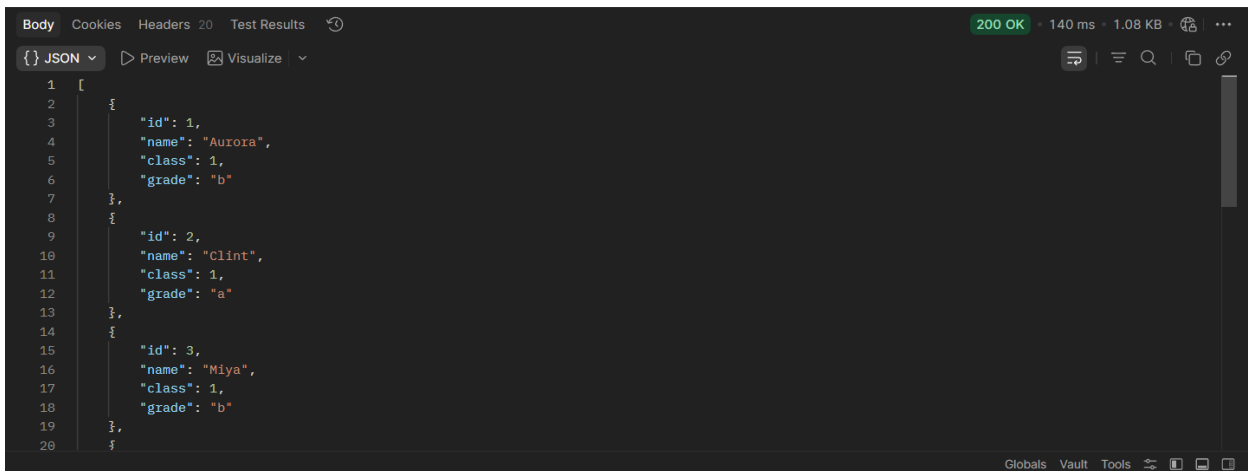


Figure 6. GET response showing 200 OK status and student records in the Body tab

Retrieving a specific student record

To retrieve a specific student information, you need to include the student ID in the URL.

Endpoint:

`https://etyucsgtbtmxlimthorwr.supabase.co/rest/v1/students?id=eq.1`

Note:

Replace **1** with the ID of the student record you want to retrieve.

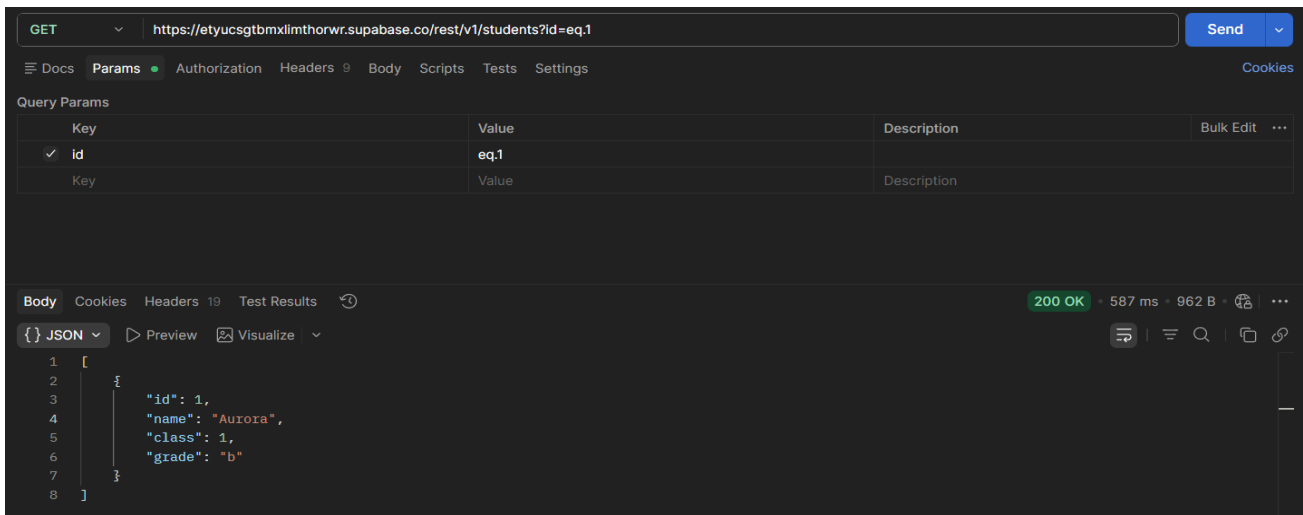


Figure 7. GET request with query parameter to retrieve a specific student record

POST method

The POST method adds a new student record. Use this method when you want to insert new student details into the database.

Endpoint: https://etyucsgtbtmxlimthorwr.supabase.co/rest/v1/students

1. Enter the endpoint URL in the URL text box.
2. Select **POST** from the HTTP method dropdown.

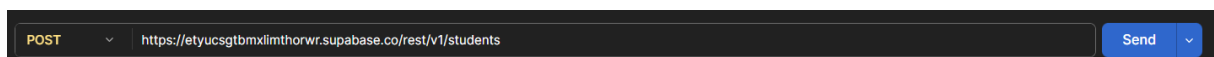


Figure 8. POST request URL in the URL text box

3. Select the **Headers** tab and add the required authentication headers as specified in the [Authentication](#) section.
4. Add the **Prefer** header

Table 3. Additional header required for POST requests

Key	Value	Description
Prefer	return=representation	Returns the created record in the response

5. Select the **Body** tab, select **raw** and set the format to **JSON** from the dropdown menu.

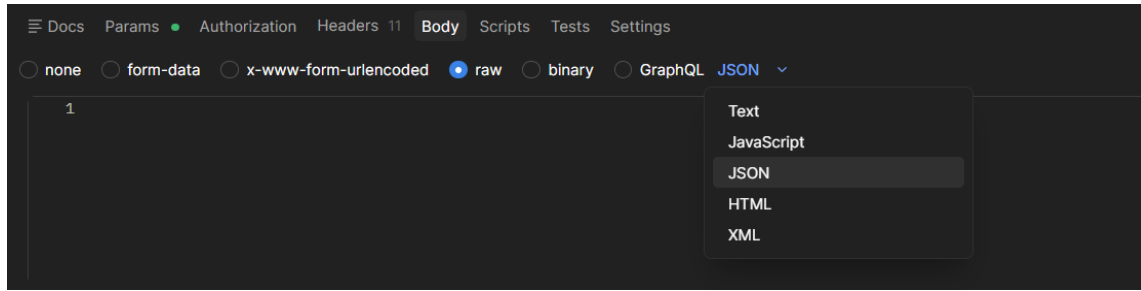


Figure 9. Body tab with raw format selected and JSON format set from the dropdown menu

6. Enter the student details in JSON format.

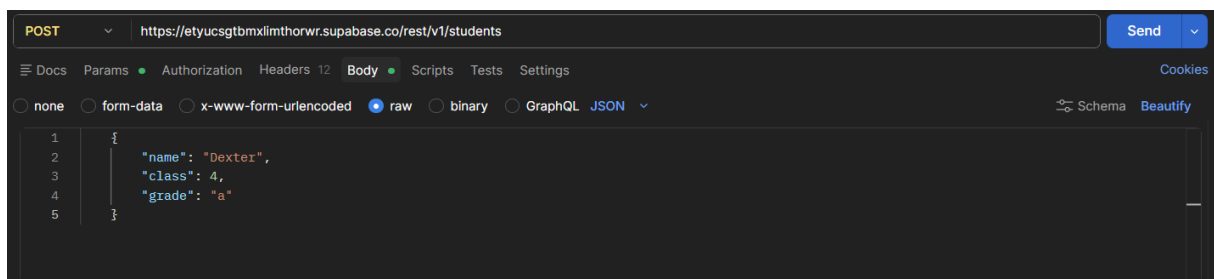


Figure 10. POST request body with new student details entered in JSON format

7. Select **Send**.
8. Verify that the response shows a **201 Created** status in the **Body** tab.



Figure 11. POST response showing 201 Created status with the new student record

Note:

The Prefer: **return=representation** header is optional. Without it, the API returns a **204 No Content** response with no data. Including it returns the newly created record in the response with a **201 Created** status. There is no need for a separate GET request to verify the record.

PATCH method

Use **PATCH** to update a student record without replacing the entire record. Use this method when you want to update specific fields of a student record.

Endpoint:

<https://etyucsgtbnxlmthorwr.supabase.co/rest/v1/students?id=eq.10>

Note:

Replace **10** with the ID of the student record you want to update.

1. Follow the same steps as the [POST method](#) section with the following exceptions:

- Select **PATCH** from the HTTP method dropdown.
- Enter the endpoint URL with the student ID as a query parameter. For more information, see [Retrieving a specific student record](#) in the **GET method** section.

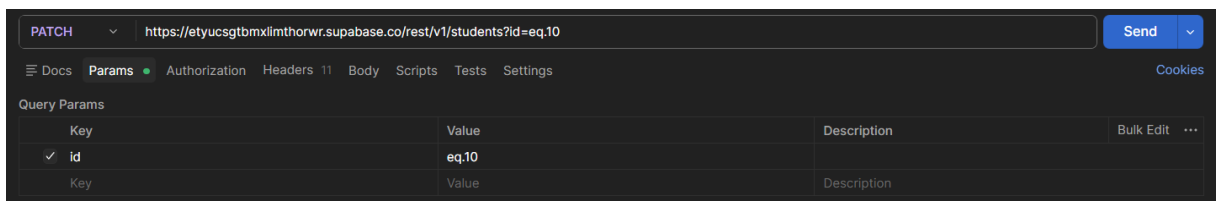


Figure 12. PATCH request with student ID query parameter in the URL text box

c. Enter only the fields you want to update in the **Body** tab.

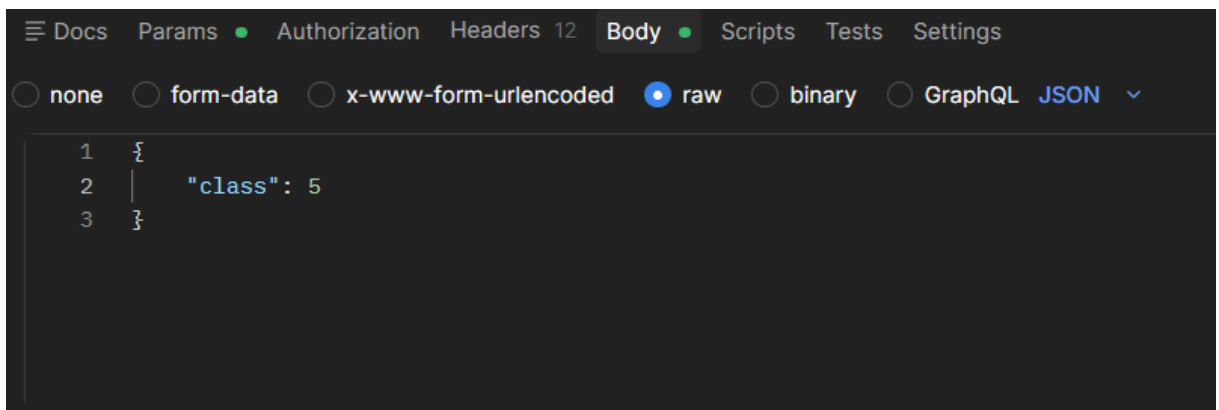
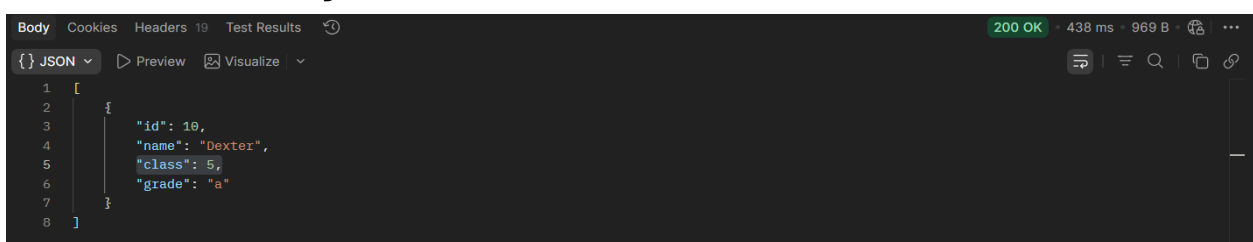


Figure 13. PATCH request Body tab with the field to update entered in JSON format

2. Verify that the response shows a **200 OK** status and the updated student record in the **Body** tab.



Note:

The response returns the complete updated record, not just the modified fields. Verify the updated values in the **Body** tab.

Warning:

Always include the **id** query parameter in the endpoint URL. Without it, the PATCH request will update **all records** in the students table.

DELETE method

Use **DELETE** to remove a student record from the students database. Use this method when you want to permanently delete a student record from the database.

Endpoint:

<https://etyucsgtbtmxmlimthorwr.supabase.co/rest/v1/students?id=eq.1>

Note:

Replace 1 with the ID of the student record you want to delete.

1. Follow the same steps as the [POST method](#) section with the following exceptions:
 - a) Select **DELETE** from the HTTP method dropdown.
 - b) Enter the endpoint URL with the student ID as a query parameter. For more information, see [Retrieving a specific student record](#) in the **GET method** section.

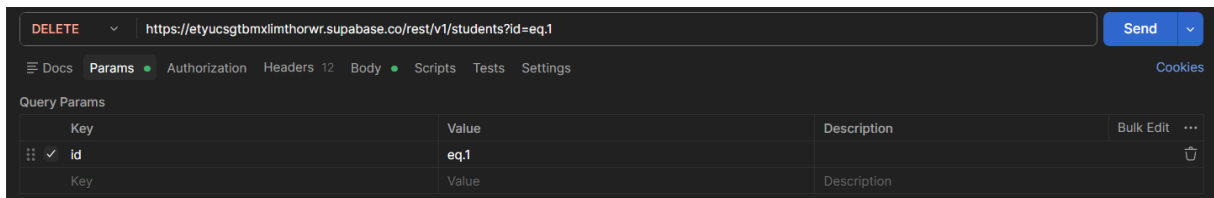


Figure 15. DELETE request with student ID query parameter in the URL text box

c) The DELETE method does not require a request **Body**. Leave the **Body** tab empty.

2. Select **Send**.

Note:

The DELETE request can return two different status codes depending on the Prefer header:

- If the **Prefer: return=representation** header is **included**, the API returns a **200 OK** status with the deleted record in the response body.



Figure 16. DELETE response showing 200 OK status when the Prefer header is included

- If the **Prefer: return=representation** header is **excluded**, the API returns a **204 No Content** status with an empty response body.



Figure 17. DELETE response showing 204 No Content status when the Prefer header is excluded

Both responses confirm that the student record has been successfully deleted.

Warning:

Always include the **id** query parameter in the endpoint URL. Without it, the DELETE request will permanently remove **all records** from the students table. This action cannot be undone.

API response codes

This section lists the HTTP status codes returned by the Student Database API. Use this section to identify and resolve errors that occur during API requests in Postman.

Success codes

Table 4. HTTP success status codes

Status Code	Method	Description
200 OK	GET, PATCH	Request was successful
201 Created	POST	Record was created successfully
204 No Content	DELETE	Record was deleted successfully

Error codes

Table 5. HTTP error status codes

Status Code	Description	Cause	Resolution
400 Bad Request	Invalid request format	Incorrect JSON body	Check the request body format
401 Unauthorized	Authentication failed	Missing or incorrect API key	Check the headers in the Authentication section
404 Not Found	Record does not exist	Wrong ID or endpoint	Check the student ID and endpoint URL
405 Method Not Allowed	HTTP method not supported	Wrong method selected	Select the correct HTTP method from the dropdown
500 Internal Server Error	Server-side error	Server error	Retry the request later

